

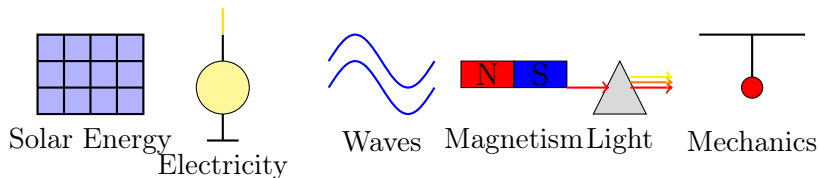
PHYSICS

SENIOR 1 TO SENIOR 2

COMPREHENSIVE NOTES

Based on the Lower Secondary Curriculum (NCDC, Uganda)

Designed for Up-Country Schools with Simple, Affordable Learning Aids



Prepared by
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Acknowledgement

I am **Mungutimudit Kelly**, a teacher of Physics and Mathematics with a Bachelor's Degree in Education. My deepest motivation in writing this work is the desire to guide and inspire fellow teachers, researchers, parents, and students to reach their highest potential. This book is written with the sole purpose of helping learners and researchers across the world – and particularly in Uganda – to develop a genuine love for Physics and to appreciate its teaching, practical experiences, and real-world applications.

I extend my sincere gratitude to all Physics, Science, and other teachers worldwide who continue to dedicate themselves to shaping future leaders, scientists, and innovators with the knowledge and vision they need to succeed.

To my lecturers – **Prof. Dennis Atibuni, Dr. Ecuru Paul, Dr. Oryema Bosco, and Dr. Nasejje Stella** of Muni University – I am forever grateful for your mentorship and guidance throughout my undergraduate journey.

To all my secondary school teachers at **Angal SS** and **Paidha SS**, thank you for laying the foundation for my education.

To my parents, **Mr. Gingyera Paul** (Teacher at Paidha SS – Zombo District) and **Atimango Florence**, my mother, I am deeply thankful for your unwavering support, both in material needs and accommodation wherever my studies took me.

To my family, especially my beloved wife **Yikparwoth Lauren Gamba**, for her love and care during my school days. I pray that the Almighty rewards you abundantly.

And to my son **Jcopo Sebastian Neville**, whom I consider a blessing from the Most High.

To colleagues, friends, and in-laws – especially **Mr. Maditrwoth Augustine, Mr. Awekonimungu Emmanuel**, and everyone we have crossed paths with – I say thank you.

Mungutimudit Kelly
2026

Dedication

This work is dedicated to:

- The Almighty God, for the gift of life, wisdom, and strength to complete this work.
- My beloved wife, **Yikparwoth Lauren Gamba**, whose love, patience, and care have been my strength during my studies.
- My son, **Jcopo Sebastian Neville**, a blessing from the Most High, who inspires me to work harder for a better future.
- My parents, **Mr. Gingyera Paul** and **Atimango Florence**, for their sacrifices, prayers, and support throughout my education.
- All teachers and learners across Uganda and the world, who continue to strive for excellence in science and education.
- Every aspiring physicist, researcher, and future leader who will use this work to grow in knowledge and wisdom.

“Education is the most powerful weapon which you can use to change the world.”

— Nelson Mandela

Introduction

Physics is the study of matter, energy, and how they interact. It is the foundation of all natural sciences and explains how the world around us works – from the falling of rain to the movement of planets, from the glow of a bulb to the power of a nuclear plant.

This book is designed for Senior 1 and Senior 2 students in Uganda, following the Lower Secondary Curriculum developed by the National Curriculum Development Centre (NCDC). It is written with a special focus on **up-country schools**, where access to expensive laboratory equipment and learning materials may be limited. The notes, activities, and demonstrations in this book use **simple, affordable, and locally available materials** to help learners understand Physics in a practical and meaningful way.

The book is structured to promote **student-centered learning, cooperation, and critical thinking**. Each topic includes:

- **Clear explanations** of key concepts in simple language.
- **Activity-based scenarios** that encourage learners to work together and explore Physics in real-life contexts.
- **Practical demonstrations** using cheap and locally available materials.
- **Diagrams and illustrations** to aid understanding.
- **Suggested answers** to activities, helping learners check their progress.
- **Video links** (where internet is available) to enhance learning through visual demonstrations.
- **Boxes for teachers and students** with clear instructions on what to do and what not to do.

The goal of this book is not just to help learners pass examinations, but to help them **appreciate Physics** as a subject that explains the world and improves lives. It is my hope that this work will inspire the next generation of Ugandan scientists, engineers, teachers, and innovators.

Whether you are a student, a teacher, a parent, or a researcher, I pray that this book will be a valuable companion in your journey of learning and discovery.

Mungutimudit Kelly
Teacher of Physics and Mathematics
2026

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